

Презентация

Лабораторная работа 5

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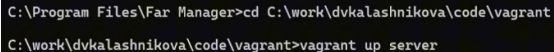
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Информация

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Получить практические навыки расширенной настройки HTTP-сервера Apache: защита соединения и поддержка PHP.

Поднимаем сервер через vagrant и убеждаемся, что он запустился.

A terminal window with a black background and white text. The first line shows the command 'cd C:\work\dvkalashnikova\code\vagrant' being executed from the 'C:\Program Files\Far Manager' directory. The second line shows the command 'vagrant up server' being executed from the directory 'C:\work\dvkalashnikova\code\vagrant'.

```
C:\Program Files\Far Manager>cd C:\work\dvkalashnikova\code\vagrant  
C:\work\dvkalashnikova\code\vagrant>vagrant up server
```

Рис. 1: Запуск сервера

Генерація сертификата

Создаём SSL-сертификат в `/etc/pki/tls/private` и копируем его в `/etc/ssl/certs`.

[illegible]

Рис. 2: Создание сертификата

Открываем в /etc/httpd/conf.d файл www.dvkalashnikova.net.conf.

```
[root@server.dvkalashnikova.net private]# cd /etc/httpd/conf.d
[root@server.dvkalashnikova.net conf.d]# nano /etc/httpd/conf.d/www.dvkalashnikova.net.conf
```

Рис. 3: Редактирование конфигурационного файла httpd

Приводим конфиг к виду с поддержкой HTTPS: два блока VirtualHost и редирект с 80 на 443.

```
GNU nano 8.1 /etc/httpd/conf.d/www.dvkalashnikova.net.conf
<VirtualHost *:80>
    ServerAdmin webmaster@dvkalashnikova.net
    DocumentRoot /var/www/html/www.dvkalashnikova.net
    ServerName www.dvkalashnikova.net
    ServerAlias www.dvkalashnikova.net
    ErrorLog logs/www.dvkalashnikova.net-error_log
    CustomLog logs/www.dvkalashnikova.net-access_log common
    RewriteEngine on
    RewriteRule ^(.*)$ https://%{HTTP_HOST}%1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
    SSLEngine on
    ServerAdmin webmaster@dvkalashnikova.net
    DocumentRoot /var/www/html/www.dvkalashnikova.net
    ServerName www.dvkalashnikova.net
    ServerAlias www.dvkalashnikova.net
    ErrorLog logs/www.dvkalashnikova.net-error_log
    CustomLog logs/www.dvkalashnikova.net-access_log common
    SSLCertificateFile /etc/pki/tls/certs/www.dvkalashnikova.net.crt
    SSLCertificateKeyFile /etc/pki/tls/private/www.dvkalashnikova.net.key
</VirtualHost>
</IfModule>
```

Рис. 4: www.dvkalashnikova.net.conf

Настройки фаервола

Разрешаем работу по https через firewall-cmd и перезапускаем httpd.

```
cockpit dhcp dhcpv6-client dns http ssh
[rootserver.dvkalashnikova.net conf.d]# %C
[rootserver.dvkalashnikova.net conf.d]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp am
aps avno-1602 avno-1800 apcussd aseqnet auditl auselstap02 bacula bacula-client bazooka-dire
ctor baross-filedemon baross-storage bb hpa bitcoin bitcoind-rpc bitcoind-testnet bitcoind-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-
iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-uni
cast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-qtcp dns-over-tls docker-registry docke
r-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp gale
a ganglia-client ganglia-master git gitea gssd grafana gre high-availability http http3 ht
tps ident imap imaps iperf2 iperf3 ipfs ipsec irc trcs iscsi-target iroha je
nkins kadmim kubeconnect kubernetes kibana klogon kpasswd kpropp kubell kube-api kube-apicurve
r kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-man
ager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubele
t kubelet-rookonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmwr l
lms-client llms-top llms-udp managesieve matrix mino memcached minecraft minecraft-s
onigds mosh mountd mpd mqtt mqtt-tls ns-wdt nsssl nurmar mysql nbd nebula need-for-speed-m
ost-wanted netbios-ns netdata-dashboard nfs nfs3 nmap-0183 nrpe ntp nut opentelemetry open
vpn ovirt-inagelo ovirt-storageconsole ovirt-vaconsole plex pmcd pmaproxy pwebapi pwebapi
s psapi ps2m postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp proxy-http
ps2link ps3netrvr ptp pulseaudio puppetmaster quassel radius radsec rdp redfs redis-sentin
el rookd rpc-bind rquotad rsh rsyslog rtsp salt-master samba samba-client samba-dc sane set
tlers-history-collection xtp xps xlmover xlp xmp xmp-subsession xmp-x xmp-xmptle xmp
tls-trap xmptrap socks spidersok-lansync sportify-sync squid snmp ssh stattd steam-lan-tr
ansfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart sv
dcp svn syncthing syncthing-gui syncthing-relay synergy sysconlan syslog syslog-ng syslog-
tls telnet tentacle terraria tfpp tile38 tinc tor-socks transmission-client turn turns ipn
p-client vdsu vnc-server vrrp wargamestar when-http when-https wireguard ws-discovery ws-di
scovery-client ws-discovery-host ws-discovery-top ws-discovery-udp wsd wsd-http wsmem ws
mans xdcop xmp-bosh xmp-client xmp-local xmp-server zabbix-agent zabbix-java-gateway z
abbix-server zabbix-trapper zabbix-web-service zero-k zerotier
[rootserver.dvkalashnikova.net conf.d]# firewall-cmd --add-service=https
success
[rootserver.dvkalashnikova.net conf.d]# firewall-cmd --add-service=https --permanent
success
[rootserver.dvkalashnikova.net conf.d]# firewall-cmd --reload
success
[rootserver.dvkalashnikova.net conf.d]# systemctl restart httpd
```

Рис. 5: Настройка фаервола

Предупреждение браузера

На клиенте открываем сайт — браузер предупреждает, что сертификат самоподписанный.

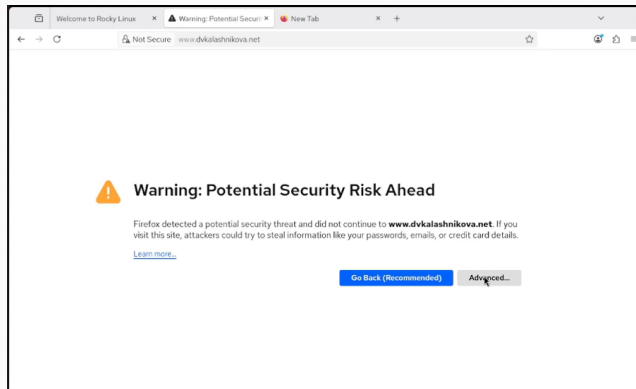


Рис. 6: Предупреждение

Подтверждаем переход и видим, что соединение идёт по https.

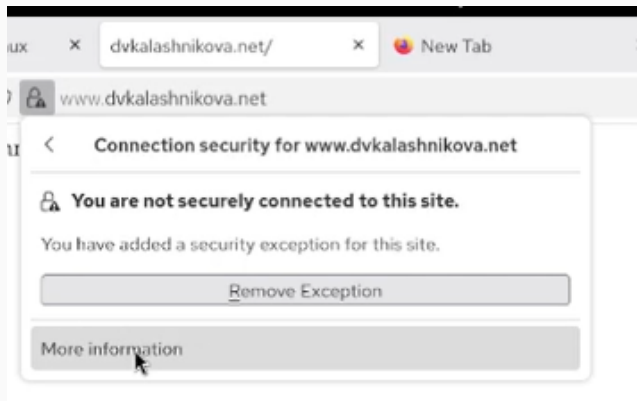


Рис. 7: Подключение к сайту

В сертификате те же данные, что мы указывали при создании.

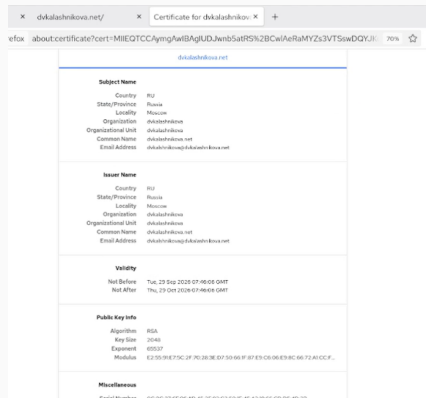


Рис. 8: Сертификат

Ставим на сервер пакет php для обработки динамических страниц.

```
[root@server.dvkalashnikova.net ~]# dnf -y install php
Last metadata expiration check: 1:27:56 ago on Tue 29 Sep 2026 08:02:14 PM UTC.
Dependencies resolved.
=====
Package                Architecture Version                Repository            Size
=====
Installing:
php                    x86_64          8.3.33-1.el10_2       appstream              7.8 k
Installing dependencies:
capstone               x86_64          5.0.1-8.el10_2        appstream              1.0 M
nginx-filesystem       noarch          2:1.26.3-6.el10_2.7   appstream              11 k
php-common             x86_64          8.3.33-1.el10_2       appstream              708 k
Installing weak dependencies:
php-cli                x86_64          8.3.33-1.el10_2       appstream              3.6 M
php-fpm                x86_64          8.3.33-1.el10_2       appstream              1.9 M
php-mbstring           x86_64          8.3.33-1.el10_2       appstream              517 k
php-opcache            x86_64          8.3.33-1.el10_2       appstream              363 k
php-pdo                x86_64          8.3.33-1.el10_2       appstream              85 k
php-xml                x86_64          8.3.33-1.el10_2       appstream              145 k

Transaction Summary
=====
Install 10 Packages

Total download size: 8.2 M
Installed size: 50 M
Downloading Packages:
[  ===                               ] --- B/s | 0 B    --:-- ETA
```

Рис. 9: Установка php

Замена index.html на index.php

В /var/www/html/www.dvkalashnikova.net создаём index.php вместо старого index.html.

```
[root@server.dvkalashnikova.net ~]# touch /var/www/html/www.dvkalashnikova.net/index.php
[root@server.dvkalashnikova.net ~]# nano /var/www/html/www.dvkalashnikova.net/index.php
[root@server.dvkalashnikova.net ~]# rm /var/www/html/www.dvkalashnikova.net/index.html
rm: remove regular file '/var/www/html/www.dvkalashnikova.net/index.html'? y
```

Рис. 10: Замена файла index.html на index.php

В index.php записываем приветственный код с использованием PHP.

A screenshot of a GNU nano 8.1 text editor window. The title bar shows the file path /var/www/html/www.dvkalashnikova.net/index.php. The editor contains the following PHP code: <?php, phpinfo();, and ?>. The cursor is positioned at the end of the third line.

```
GNU nano 8.1 /var/www/html/www.dvkalashnikova.net/index.php
<?php
phpinfo();
?>
```

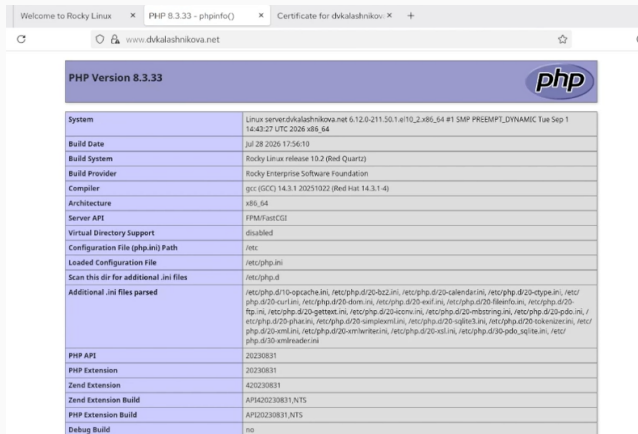
Рис. 11: Содержимое index.php

Возвращаем метки SELinux, меняем владельца /var/www на apache и перезапускаем службу.

```
[root@server.dvkalashnikova.net ~]# chown -R apache:apache /var/www  
[root@server.dvkalashnikova.net ~]# restorecon -vR /etc  
[root@server.dvkalashnikova.net ~]# restorecon -vR /var/www  
[root@server.dvkalashnikova.net ~]# systemctl restart httpd
```

Рис. 12: Исправление настроек доступа

Проверяем с клиента — сайт отдаёт страницу, сгенерированную PHP.



System	Linux server.dvkalashnikova.net 6.12.0-211.50.1.el10_2.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Sep 1 14:43:27 UTC 2026 x86_64
Build Date	Jul 28 2026 17:56:10
Build System	Rocky Linux release 10.2 (Red Quartz)
Build Provider	Rocky Enterprise Software Foundation
Compiler	gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-fpm.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-imagick.ini, /etc/php.d/20-ldap.ini, /etc/php.d/20-libsmbclient.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-readline.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlreader.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmldr.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831,NTS
PHP Extension Build	API20230831,NTS
Debug Build	no

Рис. 13: Отображение сайта

Переносим получившиеся конфиги в provision/server для vagrant.

```
[root@server.dvkalashnikova.net ~]# cd /vagrant/provision/server/http
[root@server.dvkalashnikova.net http]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autotindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.dvkalashnikova.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.dvkalashnikova.net.conf'? y
[root@server.dvkalashnikova.net http]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.dvkalashnikova.net/index.html'? y
[root@server.dvkalashnikova.net http]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.dvkalashnikova.net http]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.dvkalashnikova.net http]# cp -R /etc/pki/tls/private/www.dvkalashnikova.net.key /vagrant/provision/ser
ver/http/etc/pki/tls/private
[root@server.dvkalashnikova.net http]# cp -R /etc/pki/tls/certs/www.kalashnikova.net.crt /vagrant/provision/serve
r/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.kalashnikova.net.crt': No such file or directory
[root@server.dvkalashnikova.net http]# cp -R /etc/pki/tls/certs/www.dvkalashnikova.net.crt /vagrant/provision/ser
ver/http/etc/pki/tls/certs
[root@server.dvkalashnikova.net http]# nano /vagrant/provision/server/http.sh
[root@server.dvkalashnikova.net http]# nano /vagrant/provision/server/http.sh
[root@server.dvkalashnikova.net http]# █
```

Рис. 14: Сохранение конфигурации

Дорабатываем http.sh: он ставит php и открывает https в фаерволе.

```
GNU nano 8.1 /vagrant/provision/server/http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd

echo "copy ssl certificates"
mkdir -p /etc/pki/tls/private
mkdir -p /etc/pki/tls/certs
cp -R /vagrant/provision/server/http/etc/pki/tls/private/* /etc/pki/tls/private/
cp -R /vagrant/provision/server/http/etc/pki/tls/certs/* /etc/pki/tls/certs/
restorecon -vR /etc/pki/tls

echo "install php"
dnf -y install php
echo "configure firewall for https"
firewall-cmd --list-services
firewall-cmd --get-services
firewall-cmd --add-service=https
firewall-cmd --add-service=https --permanent
firewall-cmd --reload
echo "restart httpd"
systemctl restart httpd
```

В ходе лабораторной работы получены навыки расширенного конфигурирования HTTP-сервера Apache и организации https.

1. Apache Software Foundation. Apache HTTP Server Documentation [Электронный ресурс]. — URL: <https://httpd.apache.org/docs/> (дата обращения: 25.09.2026).
2. Apache Software Foundation. mod_ssl — SSL/TLS Encryption [Электронный ресурс]. — URL: https://httpd.apache.org/docs/2.4/mod/mod_ssl.html (дата обращения: 25.09.2026).
3. Red Hat. Documentation [Электронный ресурс]. — URL: <https://docs.redhat.com/> (дата обращения: 25.09.2026).